Lecture 7

2/4/26

We now have a feeling for Rayleigh's theory of resolution

- Apertures diffract light
- Diffraction causes the image of objects that are too close together to blur together & look like a single object

Abbe Theory of Image Formation

In many imaging modes, object diffracts/scatters light in a specimen-dependent way.

Incident (undiffracted) & diffracted light combine in image plane to create image.

(Abbe was a μscopist who worked before the advent of fluorescence μscopy)

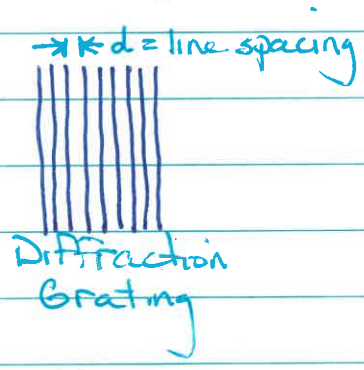
Analogy: x-ray diffraction uses diffracted x-rays to determine protein structure

- Crystallize protein
- Scatter x-rays off crystal & collect diffraction pattern
- Do snazzy math to get structure from pattern
- No x-ray μscope because no x-ray lenses

Abbe's Experiments

Abbe did a series of experiment to understand how μscopes form images & deduce the factor(s) limiting resolution

Abbe used simple objects w/ known structures in these experiments

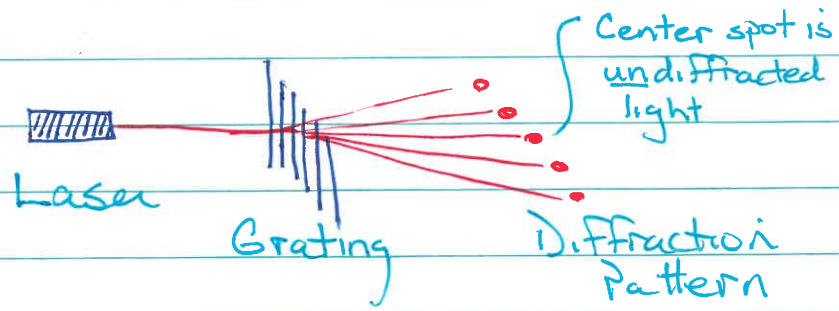


Diffraction grating = series of closely & evenly spaced lines

Gratings diffract light in an object-dependent way

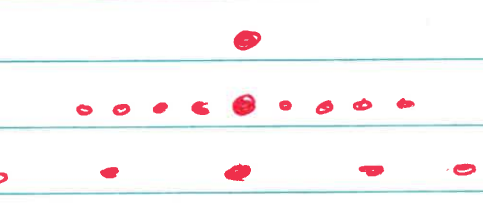
As the line spacing, d, decreases the lines become too close together to resolve

DEMO



The spacing of spots in the diffraction pattern depends on the spacing of lines in the grating

- No grating — No extra spots
- 100 lines/mm — Spots closer together
- 600 lines/mm — Spots farther apart

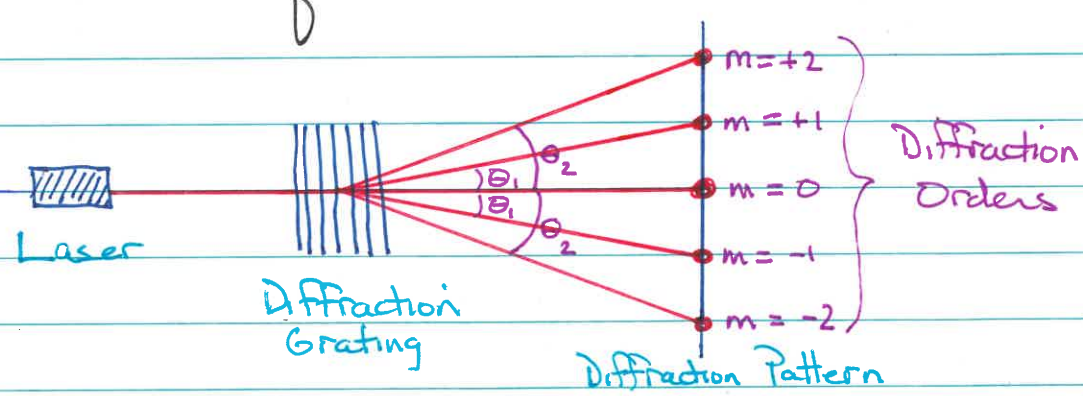


As the lines get closer together, & "d" decreases, corresponding to finer details in the object, the spots get further apart

The orientation of the spots depends on the orientation of the lines in the grating

- Vertical lines → Horizontal spots
- Horizontal lines → Vertical spots

The relationship between line spacing & spot spacing can be quantified:



How far apart are spots in pattern?

$$\boxed{m\lambda = d\sin\theta_m}$$

Calculate, for example, position of second-order spot

$$2\lambda = d\sin\theta_2 \Rightarrow \theta_2 = \arcsin\left(\frac{2\lambda}{d}\right)$$

Abbe's Thinking

Use closely spaced (small d) gratings as models for closely spaced objects or other features in specimen

$$\boxed{d_{min} = \frac{0.5\,\lambda_0}{NA}}$$

Abbe

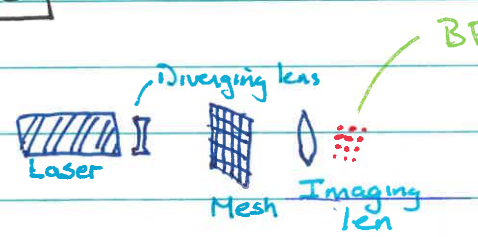
$$\boxed{d_{min} = \frac{0.61\,\lambda_0}{NA}}$$

Rayleigh

These results are very close & lead to same generalizations:

Better resolution $\begin{cases} \lambda_0 \downarrow & \text{Shorter wavelength light} \\ NA \uparrow & \text{Higher numerical aperture} \end{cases}$

DEMO



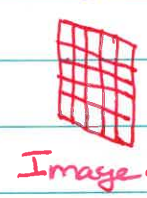
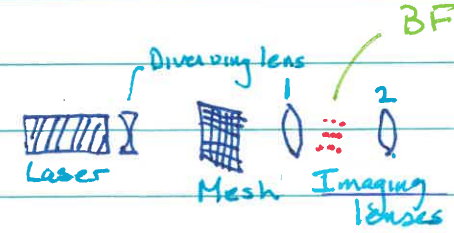


Image of mesh

Use imaging lens to form image of mesh



Use 2nd imaging lens to form image of diffraction pattern in BFP of 1st imaging lens

Image of BFP of 1st imaging lens
$\Rightarrow$ diffraction pattern

The demo uses a diverging lens to spread out the laser light before it hits the mesh.

Use 1st imaging lens to form an image of the mesh on the wall.

Add a 2nd imaging lens positioned to form an image of the BFP of the 1st imaging lens on the wall. This BFP image is a diffraction pattern of the mesh.

The wall, in each case, is the image plane.

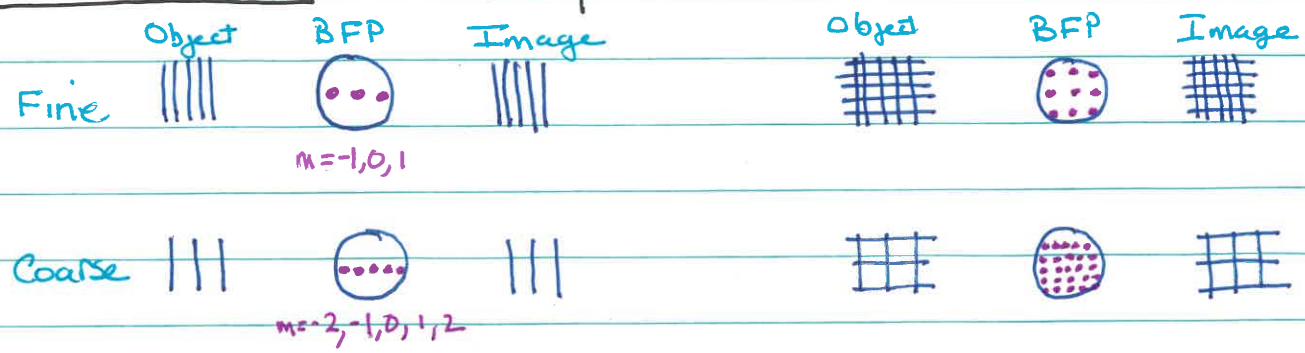
Restated Demo results:

(1) Lens forms an image in the image plane
*(2) Lens forms an object diffraction pattern in the BFP of the lens

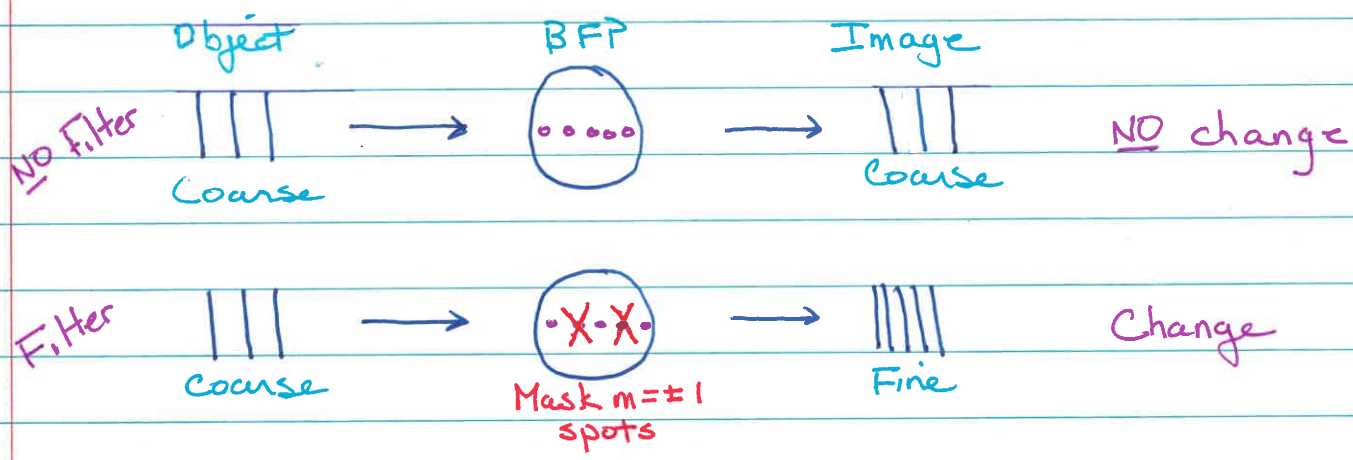
* Big surprise w/ big implications

ANIMATION

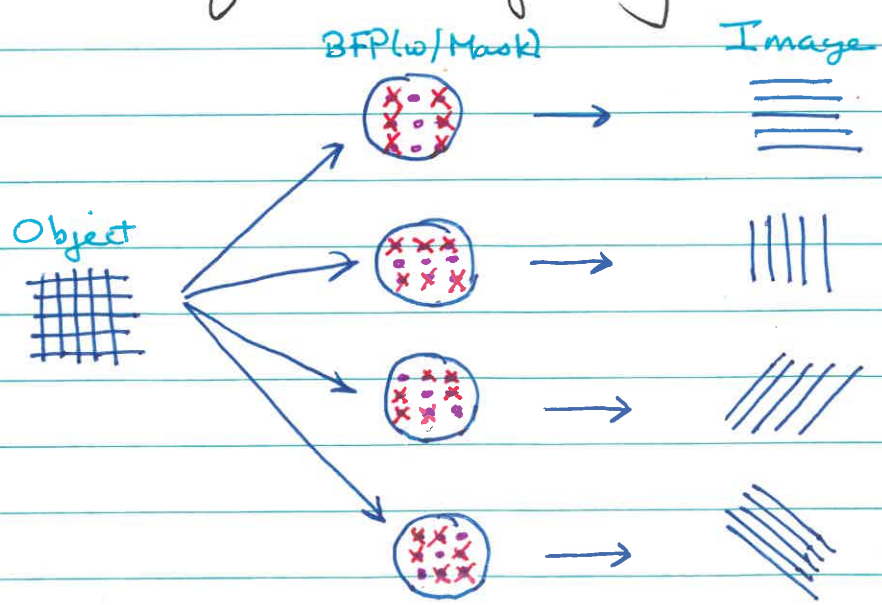
Mol Exp



Simulation can be used to explore how Abbe manipulated the diffraction pattern in the BFP of the objective & how the manipulation altered the image.



Converting the diffraction pattern of a coarse grating into the diffraction pattern of a fine grating by masking the m = ±1 spots converts the image from that of a coarse grating into that of a fine grating!



Abbe's Observations

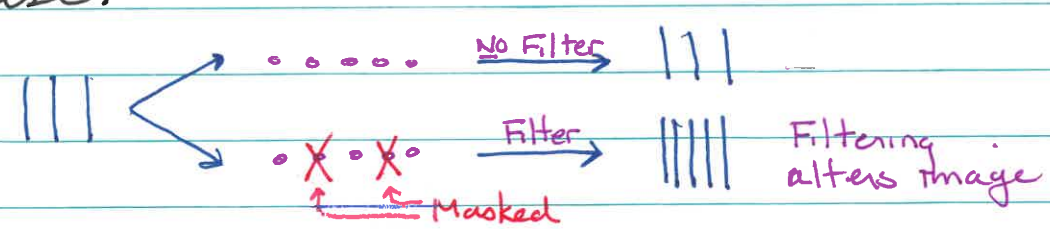
(1) See image of object in image plane

(2) See diffraction pattern of object in back focal plane (BFP)

(3) Inverse relationship between spacing of features in object & spacing of corresponding features in diffraction pattern

- Closely spaced lines in object create spots that are far apart in BFP
- Far apart lines in object create spots that are closely spaced in BFP

(4) Spatial filter the diffraction pattern by masking selected spots in BFP, altering image. This is very important because spatial filtering can be used to alter (improve) contrast.



The μscope reconstructs the image based on the diffraction pattern

(5) No image, just raw illumination, if only undiffracted ($m=0$) light passed from BFP (i.e., if mask all spots except central spot).

No specimen detail. This is when you reach the resolution limit.

Need at least one diffraction order from BFP to form an image & get resolution.

(6) Quality of image improves if more diffraction orders contribute

# of diffraction orders collected ↑ Image quality (higher resolution) ↑

(7) Quantify result for minimum resolution

$$d_{min} = \frac{0.5\lambda_0}{NA}$$

Lines more closely spaced than d_{min} are <u>NOT</u> resolvable.

Similar requirements apply to MRI
- Collect less k space info to enhance speed
- Collect more k space info to enhance quality